

Beyond Fairness:

Educational Responsibility in LLM-Personalised Learning

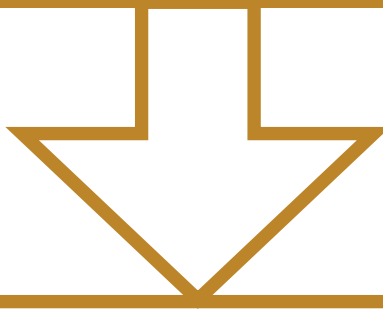
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Personalisation ≠ Fairness

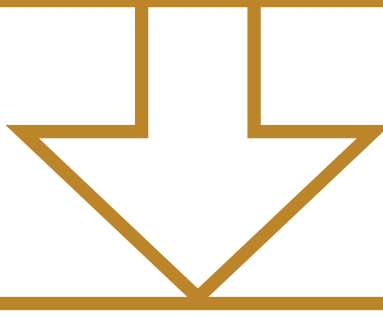
LLM-personalised learning can reproduce educational inequality when learner background cues are interpreted as social assumptions.

How the risk emerges?

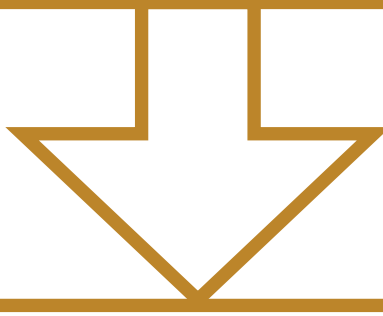
Learner background cues



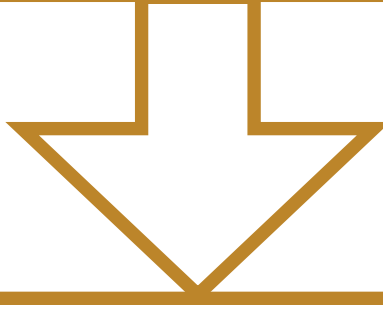
Implicit SES inference



Personalised response framing



Learner positioning & expectations



Educational consequences

This diagram shows how learner background cues may be translated into socioeconomic assumptions that shape personalised responses and their educational consequences.

Fairness vs Educational Responsibility

Technical Fairness

- Bias detection
- Output parity
- Model performance
- Post-hoc correction

VS

Educational Responsibility

- Pedagogical consequences
- Learner positioning
- Value assumptions
- Accountability

Research & Governance Agenda

- Evaluate SES assumptions in personalised responses
- Examine how learners are positioned by LLM advice
- Include educators and learners in governance design
- Move from bias correction to educational accountability

Two key challenges

Challenge 1: Deficit framing

Socioeconomic cues may be read as lower motivation, ability, or aspiration.

Challenge 2: Hidden accountability

When personalised advice shapes opportunities, responsibility becomes hard to locate.

Discussion Questions

- What kinds of learner profiling are pedagogically acceptable?
- How can SES-aware evaluation avoid deficit framing?
- Who is accountable when personalisation lowers expectations?
- How should governance include diverse educational contexts?

Takeaway

Beyond fairness, LLM-personalised learning requires responsibility for how learners are interpreted, positioned, and supported.